

claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_e2a9aaa7-6b2\agent-a5086d1ce0dec1cd4.jsonl`  
- SHA-256: `c3f998355e5232948f65b9ded8a4f915e43ea0de1fe160ac438b03433398d062`  
- Source modified: `2026-05-30T19:13:10+00:00`  
- Imported at: `2026-07-05T16:48:21+00:00`  
- project: `wf_e2a9aaa7-6b2`  
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`
```

Transcript

1. user (2026-05-30T19:10:36.658Z)

Web Searcher: reuse/prior-art ? existing parsers

Research question: "Find promising AND reliable open-source tools/libraries for parsing Indian bank statement PDFs ? especially HDFC Bank savings/transaction statements ? into structured transaction rows (date, narration, ref, value date, withdrawal, deposit, closing balance), for reuse in a LOCAL-FIRST Python personal-finance tool ("muneem").

HARD CONSTRAINTS (a tool that violates these is unusable for us):

1. Must run 100% LOCALLY. Bank statements are highly sensitive ? NO cloud APIs, NO paid SaaS (e.g. bankstatementconverter.com, docparser, LLM extraction APIs). Paid/cloud is not acceptable even as a default.
2. Must be usable from Python.
3. License MUST be permissive & commercially safe: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL or that depends on copyleft native components ? in particular Camelot/tabula's reliance on Ghostscript (AGPL) or a Java runtime. PyMuPDF (AGPL) is already banned for us.

Specifically evaluate and compare these table-extraction approaches, for each giving: license (AND license of native deps), maintenance status (latest release + activity as of 2025/2026), reliability on SEMI-RULED / partially-borderless tables (HDFC's transaction table is not fully line-ruled), and password-protected-PDF support:

- pdfplumber (we already use it; MIT)
- Camelot ? camelot-py vs the maintained "pypdf_table_extraction" fork; its Ghostscript/OpenCV deps
- tabula-py (tabula-java + JVM)
- pypdfium2-based table extraction
- any other notable local options

Also assess the user's idea: "convert the PDF table to Excel/CSV first, then parse" ? is an intermediate CSV/Excel extraction more reliable than direct coordinate/text parsing, and which LOCAL+permissive tools do PDF?CSV/Excel table conversion well?

Finally, search for any existing, maintained, permissively-licensed open-source projects specifically for parsing Indian bank statements (HDFC/ICICI/SBI/Axis) or bank statements generally, that we could reuse or learn from (GitHub/PyPI).

Deliver a ranked shortlist of the most promising + reliable + license-safe options for HDFC statement parsing in a local Python tool, each with its license (and native-dep licenses), maintenance status, strengths/weaknesses for this use case, and a concrete recommendation for muneem (reuse library X vs. hand-roll coordinate clustering with pdfplumber). Cite sources."

Your angle: **reuse/prior-art ? existing parsers** ? Hunts GitHub/PyPI for maintained, permissively-licensed Indian or generic bank-statement parsers muneem could reuse or learn from, and reveals real-world PDF?CSV/Excel approaches for Indian banks.

Search query: `open source HDFC ICICI SBI Axis bank statement parser Python GitHub pdf to csv`

transactions MIT`

Task

Use WebSearch with the query above (or a refined version). Return the top 4-6 most relevant results.

Rank by relevance to the ORIGINAL question, not just the search query. Skip obvious SEO spam/content farms.

Include a short snippet capturing why each result is relevant.

Structured output only.

2. assistant (2026-05-30T19:10:40.263Z)

I'll research this thoroughly. Let me load the WebSearch tool and run targeted searches.

3. assistant (2026-05-30T19:10:43.288Z)

I'll run several targeted searches to find existing parsers and prior art.

4. user (2026-05-30T19:10:51.135Z)

Web search results for query: "open source HDFC ICICI SBI Axis bank statement parser Python GitHub pdf to csv transactions MIT"

Links: [{"title": "GitHub - sebastienrousseau/bankstatementparser: Parse bank statements across CAMT, PAIN.001, CSV, OFX/QFX, MT940, and PDFs (digital + scanned) into unified Transaction models. Deterministic ISO 20022 parsers, LLM fallback for PDFs, vision for scans, balance verification, categorization, and interactive review mode. · GitHub", "url": "https://github.com/sebastienrousseau/bankstatementparser"}, {"title": "Bank Statement Parser: Parse 7 Formats in Python, 100% Local", "url": "https://bankstatementparser.com/"}, {"title": "GitHub - trakanom/Bank-Statement-Parser-PDF-to-CSV: Converts bank statements to pandas tables and exports to .csv. Modular design with validation, calibrated for Wells Fargo. · GitHub", "url": "https://github.com/trakanom/Bank-Statement-Parser-PDF-to-CSV"}, {"title": "GitHub - felgru/bank-statement-parser: scripts to import bank statement PDFs into hledger files · GitHub", "url": "https://github.com/felgru/bank-statement-parser"}, {"title": "GitHub - seahyc/bank_statement_parser: Parse OCBC or DBS banks statements into individual transactions", "url": "https://github.com/seahyc/bank_statement_parser"}, {"title": "GitHub - the-tech-project/statement-parser: A Python based project to parse bank statements. · GitHub", "url": "https://github.com/the-tech-project/statement-parser"}, {"title": "Bank Statement Parser using Python - Bhavesh Bhatt", "url": "https://bhattbhavesh91.github.io/python/bank-statement-parser/"}, {"title": "bank-statements · GitHub Topics · GitHub", "url": "https://github.com/topics/bank-statements"}, {"title": "bank-statement-parser · GitHub Topics · GitHub", "url": "https://github.com/topics/bank-statement-parser"}, {"title": "bank-statement · GitHub Topics · GitHub", "url": "https://github.com/topics/bank-statement"}]

I found several open-source Python bank statement parser projects on GitHub. Here are the key findings:

Primary Results:

1. Bank Statement Parser (sebastienrousseau)

An open-source Python library to parse CAMT.053, PAIN.001, CSV, OFX, QFX, MT940, and PDF bank statements. It parses statements into structured pandas DataFrames, with all processing running locally ? deterministic output, automatic PII redaction, and an optional hybrid PDF pipeline that routes through local LLMs when needed. It's Apache 2.0 licensed (note: not MIT, but still permissive open-source).

GitHub: <https://github.com/sebastienrousseau/bankstatementparser>

2. Bank Statement Parser PDF-to-CSV (trakanom)

This project is released under the MIT License. It's designed to convert bank statements to pandas tables and export to CSV format.

GitHub: <https://github.com/trakanom/Bank-Statement-Parser-PDF-to-CSV>

Additional Options:

- A Python tool to parse bank statements from PDF files, specifically designed for OCBC and DBS statements.
- Monopoly is a Python library & CLI that converts bank statement PDFs to CSV.

Note: The search results don't show projects specifically tailored for Indian banks (HDFC, ICICI, SBI, Axis). You may need to explore the GitHub repositories directly or consider creating custom parsing rules for these banks' specific statement formats within one of these parser frameworks.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

5. user (2026-05-30T19:10:51.732Z)

Web search results for query: "GitHub Indian bank statement PDF parser pdfplumber HDFC transactions library"

Links: [{"title": "Building a PDF Parser for HDFC Bank Statements: From 165 Pages to CSV in Minutes - DEV Community", "url": "https://dev.to/vishwaraja_pathivishwa/building-a-pdf-parser-for-hdfc-bank-statements-from-165-pages-to-csv-in-minutes-34c6"}, {"title": "bank-statement - GitHub Topics", "url": "https://github.com/topics/bank-statement?l=python&o=asc&s=forks"}, {"title": "GitHub - xaneem/hdfc-credit-card-statement-parser: Extracts information from an HDFC Bank Credit Card Statement (India)", "url": "https://github.com/xaneem/hdfc-credit-card-statement-parser"}, {"title": "GitHub - MalayPalace/Bank-Statement-Utility: Parses Bank's Statement/Transaction file, Stores and Process it. Can be further used for expense tracking and analytics.", "url": "https://github.com/MalayPalace/Bank-Statement-Utility"}, {"title": "GitHub - J-sephB-lt-n/pdf-bank-statement-parser: Tool for converting First National Bank (FNB) bank statement PDFs into useful structured data", "url": "https://github.com/J-sephB-lt-n/pdf-bank-statement-parser"}, {"title": "GitHub - sebastienrousseau/bankstatementparser: Parse bank statements across CAMT, PAIN.001, CSV, OFX/QFX, MT940, and PDFs (digital + scanned) into unified Transaction models. Deterministic ISO 20022 parsers, LLM fallback for PDFs, vision for scans, balance verification, categorization, and interactive review mode.", "url": "https://github.com/sebastienrousseau/bankstatementparser"}, {"title": "GitHub - joeirimpan/hdfc-cc-parser-rs: Parse hdfc cc statements into csv (infinia 2025 september+)", "url": "https://github.com/joeirimpan/hdfc-cc-parser-rs"}, {"title": "bank-statements - GitHub Topics", "url": "https://github.com/topics/bank-statements"}, {"title": "hdfc-cc-parser-rs/README.md at master · joeirimpan/hdfc-cc-parser-rs", "url": "https://github.com/joeirimpan/hdfc-cc-parser-rs/blob/master/README.md"}]

Here are the search results for Indian bank statement PDF parsers using pdfplumber and related libraries:

Key Findings

Popular HDFC Bank Statement Parsers:

- hdfc-pdf-converter** - A Python tool that automatically extracts all transactions from HDFC Bank PDF statements and converts them to CSV format with intelligent categorization. It uses core dependencies including camelot-py, pandas, PyPDF2, and pdfplumber for text extraction.
- hdfc-credit-card-statement-parser** - This tool extracts information from HDFC Bank Credit Card statements to help copy structured data out of your statement. It extracts both Indian and Foreign transactions, and for Foreign transactions, the Forex Amount and Forex Rate is also extracted.
- hdfc-cc-parser-rs** - A Rust tool that parses and extracts transaction data from HDFC Bank credit card statements to CSV format.

****Multi-Bank Solutions:****

An open-source Python tool to parse credit card PDF statements from Indian banks (HDFC, ICICI, SBI, Axis + 5 more) into structured data with offline, privacy-first, Streamlit dashboard.

Bank Statement Utility supports Indian banks and has created various configurable parsers which can read and store bank statement data to database.

****Other Comprehensive Solutions:****

The bankstatementparser library offers options to add higher-fidelity table extraction with pdfplumber via the 'hybrid-plus' install variant.

These tools provide various approaches to extract and process Indian bank statements, with most supporting HDFC and other major Indian banks.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-05-30T19:10:54.362Z)

Strong leads. Let me dig deeper on the Indian-specific multi-bank tool, the HDFC parsers, and the table-extraction library landscape (Camelot fork, pypdfium2).

7. user (2026-05-30T19:11:02.122Z)

Web search results for query: "pypdf_table_extraction Camelot fork Ghostscript license maintained 2025"

Links: [{"title": "pypdf-table-extraction · PyPI", "url": "https://pypi.org/project/pypdf-table-extraction/"}, {"title": "camelot-py · PyPI", "url": "https://pypi.org/project/camelot-py/"}, {"title": "GitHub - py-pdf/pypdf_table_extraction: A Python library to extract tabular data from PDFs · GitHub", "url": "https://github.com/py-pdf/pypdf_table_extraction"}, {"title": "Integration of Camelot in py-pdf · py-pdf/pypdf · Discussion #2466", "url": "https://github.com/py-pdf/pypdf/discussions/2466"}, {"title": "Releases · py-pdf/pypdf_table_extraction", "url": "https://github.com/py-pdf/pypdf_table_extraction/releases"}, {"title": "Introduction ? pypdf_table_extraction 1.0.2 documentation", "url": "https://pypdf-table-extraction.readthedocs.io/en/latest/user/intro.html"}, {"title": "5 Python open-source tools to extract text and tabular data from PDF Files | Towards Data Science", "url": "https://towardsdatascience.com/5-python-open-source-tools-to-extract-text-and-tabular-data-from-pdf-files-9359cdde6b56/"}, {"title": "Comparing 6 Frameworks for Rule-based PDF parsing", "url": "https://www.ai-bites.net/comparing-6-frameworks-for-rule-based-pdf-parsing/"}, {"title": "ModuleNotFoundError: No module named 'ghostscript', but ghostscript installed · Issue #14 · py-pdf/pypdf_table_extraction", "url": "https://github.com/py-pdf/pypdf_table_extraction/issues/14"}, {"title": "GitHub - rajatvajpayee/table-extraction-from-PDF: Objective of this project is to extract tables and its cells from a PDF using python library camelot.", "url": "https://github.com/rajatvajpayee/table-extraction-from-PDF"}]

Based on the search results, here's what I found about pypdf_table_extraction:

Overview

pypdf_table_extraction is the community maintained fork of Camelot. pypdf_table_extraction (formerly known as Camelot) is a Python library that can help you extract tables from PDFs.

License and Dependencies

This project is licensed under the MIT License. The library depends on Ghostscript: After installing the dependencies (tk and ghostscript), you can also just use pip to install pypdf_table_extraction.

Maintenance Status as of 2025

However, it's important to note that the repository was archived by the owner on Apr 11, 2025 and it is now read-only. Additionally, the PyPI page indicates "DEPRECATED - Please use camelot-py instead."

Current Recommended Alternative

The recommended alternative going forward is camelot-py. camelot-py was released Aug 10, 2025 and is described as "PDF Table Extraction for Humans." It is licensed under the MIT License and is marked as Production/Stable.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. user (2026-05-30T19:11:02.240Z)

Web search results for query: "open source Python parse credit card PDF statements Indian banks HDFC ICICI SBI Axis offline privacy Streamlit GitHub"

Links: [{"title": "bank-statement · GitHub Topics · GitHub", "url": "https://github.com/topics/bank-statement?l=python&o=asc&s=forks"}, {"title": "pdf-statement-reader · PyPI", "url": "https://pypi.org/project/pdf-statement-reader/"}, {"title": "GitHub - xaneem/hdfc-credit-card-statement-parser: Extracts information from an HDFC Bank Credit Card Statement (India)", "url": "https://github.com/xaneem/hdfc-credit-card-statement-parser"}, {"title": "GitHub - GhufraBarcha/BankStatement-Data-Extractor: This repository contains a Streamlit application powered by GPT, designed to extract data from bank statements in PDF format. The app utilizes the ChatGPT API to read and comprehend the bank statement, providing essential information such as the highest credit and debit amounts, total balance, customer name, account number, and more. · GitHub", "url": "https://github.com/GhufraBarcha/BankStatement-Data-Extractor"}, {"title": "bank-statements · GitHub Topics · GitHub", "url": "https://github.com/topics/bank-statements"}, {"title": "GitHub - azf99/bank-statement-analysis: Extract useful insights from PDF Bank Statements(Indian Banks) using python automation · GitHub", "url": "https://github.com/azf99/bank-statement-analysis"}, {"title": "Python Scripting for PDF data extraction. Example with Credit Card Statements | by ProjectToPython | Medium", "url": "https://medium.com/@projectopython/python-scripting-for-pdf-data-extraction-example-with-credit-card-statements-c289d40518f1"}, {"title": "GitHub - electrovir/statement-parser: Parse bank and credit card statements · GitHub", "url": "https://github.com/electrovir/statement-parser"}, {"title": "GitHub - MalayPalace/Bank-Statement-Utility: Parses Bank's Statement/Transaction file, Stores and Process it. Can be further used for expense tracking and analytics.", "url": "https://github.com/MalayPalace/Bank-Statement-Utility"}]

Based on the search results, here are the key findings for open-source Python tools to parse credit card PDF statements from Indian banks:

Notable Projects

An open-source Python tool is available that parses credit card PDF statements from Indian banks (HDFC, ICICI, SBI, Axis and 5 more) into structured data with an offline, privacy-first Streamlit dashboard. This project appears to be called "stmtforge" based on the installation reference found.

Specific Indian Bank Parsers

1. **HDFC-Specific Parser**: There's a dedicated HDFC credit card statement parser that extracts both Indian and Foreign transactions, including Forex Amount and Forex Rate for foreign transactions.
2. **Multi-Bank Utility**: A Bank Statement Utility supports multiple Indian banks (HDFC, KOTAK, SBI, BOB, IDBI) for saving, current, and credit card accounts, with functionality to export

transactions to CSV files.

3. ****General Bank Statement Analysis****: There's a project to extract useful insights from PDF Bank Statements for Indian Banks using Python automation.

Streamlit-Based Solutions

A Streamlit application exists that uses ChatGPT API to extract data from bank statement PDFs, providing information such as highest credit and debit amounts, total balance, customer name, and account number. However, this approach requires API integration rather than being fully offline.

These tools are available on GitHub and can be installed via pip, making them accessible for local, privacy-focused processing of Indian bank statements.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-05-30T19:11:07.487Z)

This is a key finding: the `pypdf\_table\_extraction` fork was archived/deprecated in April 2025, and `camelot-py` was revived (new release August 2025). Both depend on Ghostscript. Let me verify Ghostscript's license situation and the Camelot dependency, plus dig into the most promising Indian-specific tools (stmtforge, xaneem HDFC parser, the DEV.to HDFC pdfplumber article).

10. user (2026-05-30T19:11:13.571Z)

Web search results for query: "stmtforge Indian bank credit card statement parser PyPI GitHub license offline"

Links: [{"title": "bank-statement · GitHub Topics · GitHub", "url": "https://github.com/topics/bank-statement?l=python&o=asc&s=forks"}, {"title": "bankstatementparser · PyPI", "url": "https://pypi.org/project/bankstatementparser/"}, {"title": "pdf-statement-reader · PyPI", "url": "https://pypi.org/project/pdf-statement-reader/"}, {"title": "GitHub - xaneem/hdfc-credit-card-statement-parser: Extracts information from an HDFC Bank Credit Card Statement (India)", "url": "https://github.com/xaneem/hdfc-credit-card-statement-parser"}, {"title": "statementsparser · PyPI", "url": "https://pypi.org/project/statementsparser/"}, {"title": "GitHub - electrovir/statement-parser: Parse bank and credit card statements · GitHub", "url": "https://github.com/electrovir/statement-parser"}, {"title": "bout · PyPI", "url": "https://pypi.org/project/bout/"}, {"title": "GitHub - MalayPalace/Bank-Statement-Utility: Parses Bank's Statement/Transaction file, Stores and Process it. Can be further used for expense tracking and analytics.", "url": "https://github.com/MalayPalace/Bank-Statement-Utility"}, {"title": "bankstatementparser", "url": "https://pydigger.com/pypi/bankstatementparser"}, {"title": "statement-parser · PyPI", "url": "https://pypi.org/project/statement-parser/"}]

Based on the search results, I found information about ****stmtforge****:

Stmtforge is an open-source Python tool to parse credit card PDF statements from Indian banks (HDFC, ICICI, SBI, Axis + 5 more) into structured data. It is offline and privacy-first with a Streamlit dashboard. It can be installed via `pip install stmtforge`.

The search results show that stmtforge is listed on PyPI and GitHub, and it's designed specifically for parsing Indian bank credit card statements with support for multiple major Indian banks. The tool emphasizes privacy and offline functionality, meaning data doesn't leave your infrastructure during processing.

However, the search results don't explicitly state the specific license type for stmtforge in the snippets shown. To get more detailed information about the license or GitHub repository specifics, you may want to visit the PyPI package page directly or check the GitHub repository for stmtforge.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

11. user (2026-05-30T19:11:16.424Z)

Web search results for query: "Camelot Ghostscript AGPL dependency optional lattice stream OpenCV license bank statement"

Links: [{"title": "On license compatibility · Issue #233 · camelot-dev/camelot", "url": "https://github.com/camelot-dev/camelot/issues/233"}, {"title": "Remove dependency on ghostscript and opencv · Issue #13 · camelot-dev/camelot", "url": "https://github.com/camelot-dev/camelot/issues/13"}, {"title": "Question about Ghostscript licensing - FLOSS - DIAL Community Hub", "url": "https://hub.osc.dial.community/t/question-about-ghostscript-licensing/2409"}, {"title": "Convert HDFC Bank Credit Card statements from PDF to Excel · GitHub", "url": "https://gist.github.com/nikhilweee/24cae428f68c153afda495dc17ef43d6"}, {"title": "Installation of dependencies ? Camelot 2.0.0rc1 documentation", "url": "https://camelot-py.readthedocs.io/en/latest/user/install-deps.html"}, {"title": "'Please make sure that Ghostscript is installed' error when it is already successfully installed · Issue #184 · atlanhq/camelot", "url": "https://github.com/atlanhq/camelot/issues/184"}, {"title": "Unknown GhostScript error after successful camelot installation", "url": "https://lightrun.com/answers/camelot-dev-camelot-unknown-ghostscript-error-after-successful-camelot-installation"}, {"title": "Advanced Usage ? Camelot 1.0.9 documentation", "url": "https://camelot-py.readthedocs.io/en/latest/user/advanced.html"}, {"title": "Ghostscript : Licensing", "url": "https://www.ghostscript.com/licensing/index.html"}]

Based on the search results, here's what I found about your query:

Camelot and Ghostscript AGPL Dependency

Camelot uses ghostscript (a GPLv3 licensed project, which is a wrapper on top of ghostscript, an AGPL licensed project) to convert PDFs to PNGs in order to do line recognition and table detection in the Lattice flavor.

Making Dependencies Optional

As of v1.0.0 ghostscript is replaced by pdfium as the default image conversion backend. The other image conversion backends can still be used and are now optional to install. The optional dependency Ghostscript can be installed using your system's package manager or by running their installer.

Lattice, Stream, and Alternative Methods

The search results include a practical example of extracting bank statements. One script combines the lattice method to deterministically find table boundaries, and then uses the stream method to extract the table structure from within the boundary.

OpenCV License Considerations

The results also mention discussions about OpenCV as a dependency. Camelot uses adaptive threshold and morphological transformations from opencv.

License Compatibility Issues

When Camelot was created, it used ghostscript because it was the best solution, but the developers didn't know about how Camelot's license (MIT) is incompatible with ghostscript's license (GPL and AGPL).

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

12. assistant (2026-05-30T19:11:21.937Z)

Excellent ? critical license clarification: ****Camelot v1.0.0+ replaced Ghostscript with pdfium as the default backend, making Ghostscript optional.**** This significantly changes the license calculus. Let me verify the most promising Indian-specific tools directly via WebFetch, and confirm the camelot-py current status and the HDFC pdfplumber prior-art article.

13. assistant (2026-05-30T19:11:24.937Z)

I'll fetch the key repos to verify license, maintenance, and approach. Running these in parallel.

14. user (2026-05-30T19:11:30.872Z)

I cannot answer any of your questions about stmtforge based on the provided content. The web page only displays an error message stating that "A required part of this site couldn't load" due to potential browser extensions, network issues, or browser settings.

To obtain information about stmtforge's license, supported banks, statement types, PDF parsing library, password protection capabilities, version details, and offline functionality, you would need to access the actual stmtforge documentation or repository, as this error page contains no such technical information.

15. user (2026-05-30T19:11:32.237Z)

HDFC Credit Card Statement Parser Overview

****License:**** MIT license

****PDF Parsing Library:**** The webpage doesn't explicitly name the library used. However, the `requirements.txt` file is referenced but not shown in the content provided, so I cannot definitively identify which PDF parsing tool is utilized.

****Password-Protected PDFs:**** Yes, the project handles encrypted statements. The instructions state: "Update the password in `run.sh`" before execution, indicating password support is built in.

****Last Update:**** The repository shows "8 Commits" on the main branch, but specific commit dates aren't visible in this content. No release versions are published.

****Statement Type:**** This tool is specifically designed for "HDFC Bank Credit Card statements" rather than savings or general bank accounts. The project explicitly extracts credit card transaction data.

****Extraction Method:**** The parser identifies and processes transaction tables from the PDF, extracting details including:

- Transaction amounts (both Indian and foreign transactions)
- Forex rates for international purchases
- Credit/Debit indicators
- Calculated totals for verification

The tool outputs structured CSV files suitable for financial tracking and reconciliation.

16. user (2026-05-30T19:11:36.356Z)

HDFC Bank Statement Parser Analysis

****Libraries Used:****

The article explicitly mentions three PDF parsing libraries: "camelot-py[cv]" for table extraction, "PyPDF2" for PDF processing, and "pdfplumber" for text extraction. All three are employed in the solution.

****Table Structure:****

The article does not specify whether HDFC's transaction tables are fully line-ruled, borderless, or semi-ruled. This detail about the PDF formatting is absent from the content.

****Extraction Approach:****

The tool handles transaction rows by identifying multiple columns containing date information, narration content, and monetary amounts. Notably, it addresses a significant challenge: "multi-line narrations" where transaction descriptions span multiple rows. The parsing logic treats everything between the date and the final five columns as narration content.

****GitHub Repository & License:****

The article links to the GitHub repository at "https://github.com/vishwaraja/hdfc-pdf-converter" and emphasizes it is "completely open source," though the specific license type is not mentioned in this content.

****Password-Protected PDFs:****

There is no mention of handling password-protected PDFs in the article.

17. assistant (2026-05-30T19:11:41.453Z)

Good progress. Let me get the stmtforge details from its GitHub repo (PyPI failed), confirm the current camelot-py license/maintenance and its pdfium default, and check the MalayPalace multi-bank Indian utility (which is config-driven for HDFC/SBI/Kotak).

18. user (2026-05-30T19:11:47.117Z)

Web search results for query: "stmtforge github repository Indian bank statement parser pdfplumber license"

Links: [{"title": "bank-statement · GitHub Topics · GitHub", "url": "https://github.com/topics/bank-statement?l=python&o=asc&s=forks"}, {"title": "GitHub - sebastienrousseau/bankstatementparser: Parse bank statements across CAMT, PAIN.001, CSV, OFX/QFX, MT940, and PDFs (digital + scanned) into unified Transaction models. Deterministic ISO 20022 parsers, LLM fallback for PDFs, vision for scans, balance verification, categorization, and interactive review mode. · GitHub", "url": "https://github.com/sebastienrousseau/bankstatementparser"}, {"title": "GitHub - MalayPalace/Bank-Statement-Utility: Parses Bank's Statement/Transaction file, Stores and Process it. Can be further used for expense tracking and analytics.", "url": "https://github.com/MalayPalace/Bank-Statement-Utility"}, {"title": "GitHub - J-sephB-lt-n/pdf-bank-statement-parser: Tool for converting First National Bank (FNB) bank statement PDFs into useful structured data · GitHub", "url": "https://github.com/J-sephB-lt-n/pdf-bank-statement-parser"}, {"title": "bank-statement-parser · GitHub Topics · GitHub", "url": "https://github.com/topics/bank-statement-parser"}, {"title": "GitHub - Akshat11111/BANK-STATEMENT-ANALYSIS: A app for analyzing bank statements, categorizing expenses, visualizing spending trends, and determining loan eligibility using machine learning.", "url": "https://github.com/Akshat11111/BANK-STATEMENT-ANALYSIS"}, {"title": "bankstatementparser · PyPI", "url": "https://pypi.org/project/bankstatementparser/"}, {"title": "GitHub - epsilonstar-02/Bank-Statement-Extractor · GitHub", "url": "https://github.com/epsilonstar-02/Bank-Statement-Extractor"}, {"title": "GitHub - felgru/bank-statement-parser: scripts to import bank statement PDFs into hledger files · GitHub", "url": "https://github.com/felgru/bank-statement-parser"}]

Based on the search results, I found information about stmtforge. Stmtforge is an open-source Python tool to parse credit card PDF statements from Indian banks (HDFC, ICICI, SBI, Axis + 5 more) into structured data. It is offline and privacy-first with a Streamlit dashboard.

The search results indicate that stmtforge uses pdfplumber as one of its dependencies for PDF processing, as it's mentioned among the related Python libraries for bank statement parsing tasks. However, the search results don't contain specific information about the GitHub repository URL for stmtforge or details about its license.

If you need more specific information about the stmtforge repository URL or its exact license, you may want to search for "stmtforge PyPI" or visit the repository directly on GitHub.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

19. user (2026-05-30T19:11:49.264Z)

Camelot-py Overview

****License:**** The project is licensed under the MIT License.

****Default Backend:**** As of the current version, "Camelot's default image-conversion backend is pdfium, which ships as a wheel ? so a plain install needs no system dependencies." Ghostscript is now optional.

****Latest Release:**** Version 1.0.9 was released on August 10, 2025, indicating the project is actively maintained.

****Java Runtime:**** Not required. The library is written in Python.

****Dependencies:**** The documentation mentions OpenCV for line detection (in the `engine="combined"` feature) and pdfium as the bundled backend. Optional dependencies include PyTorch for the `[ml]` flavor and OCR libraries for the `[ocr]` flavor.

****Password-Protected PDFs:**** The provided content does not address handling of password-protected PDFs.

****Active Maintenance:**** Yes?the latest release in 2025 and ongoing GitHub activity (55 open issues, 1 pull request) suggest the project is currently maintained.

20. user (2026-05-30T19:11:50.452Z)

Bank Statement Utility Analysis

****License:**** MIT

****Language:**** Python (100%)

****Supported Indian Banks:****

- Savings/Current: HDFC, Kotak, SBI, Bank of Baroda, IDBI, SVC
- Credit Cards: Kotak, SBI, Yes Bank

****Architecture:**** Config-driven. The system uses "configurable parser which can read and store bank statement data" with bank-specific file format requirements (txt, csv, xls, pdf).

****PDF Library:**** The project uses `pypdf` for PDF extraction, as listed in its dependencies.

****Statement Types:**** Handles both savings/current accounts and credit card statements. Different download formats are required per bank and account type.

****Last Updated:**** May 11, 2026 (v2.1.2 release date shown)

****Key Features:**** The utility can process statements, store data to Cassandra, verify transactions by comparing closing balances, and export transaction data to CSV.

21. assistant (2026-05-30T19:11:54.112Z)

I now have strong, verified findings. Two more quick fetches: confirm stmtforge's actual GitHub repo/license (it's the most directly-relevant Indian multi-bank tool), and verify pdfplumber's license/capability for the comparison. Let me also confirm the sebastienrousseau bankstatementparser license since it ranked highly.

22. user (2026-05-30T19:12:02.328Z)

Bank Statement Parser Repositories

stmtforge (Indian Banks)

Based on the webpage content, **stmtforge** is referenced in the Card-Statement-Analyser repository:

Repository: [madhav921/Card-Statement-Analyser](https://github.com/madhav921/Card-Statement-Analyser)

Description: The tool can be installed via `pip install stmtforge` and is described as an "Open-source Python tool to parse credit card PDF statements from Indian banks (HDFC, ICICI, SBI, Axis + 5 more) into structured data."

Key Features: Offline, privacy-first approach with a Streamlit dashboard interface.

License: Not explicitly stated in the provided content.

Other Indian Bank Statement Parsers

The webpage lists additional relevant repositories:

1. **Bank-Statement-Analyser** (ARAVINDHRAJA123) - Converts HDFC bank statement PDFs to Excel with spending categories
2. **personal-finance-tracker** (chinmaymisra) - Handles Axis Bank statements with React/FastAPI stack

Note: Specific license information for these repositories is not provided in the source material.

23. user (2026-05-30T19:12:03.766Z)

BankStatementParser: Key Details

License: Apache Software License 2.0, per the PyPI page stating "Apache License 2.0. See LICENSE."

Offline Capability: The library operates entirely locally by default. The documentation emphasizes: "No. Zero network calls. XML parsers enforce `no\_network=True`. No cloud, no telemetry." Ollama (a local LLM engine) is the default backend for PDF processing; cloud providers are optional alternatives.

Indian Bank Support: The documentation does not mention Indian banks specifically. It focuses on ISO 20022 standards (CAMT, PAIN formats) and generic file types (CSV, OFX, MT940).

PDF Table Extraction: The hybrid pipeline uses `pypdf` for text extraction from digital PDFs. For enhanced table handling, an optional extra (`hybrid-plus`) adds `pdfplumber`. Scanned PDFs route through multimodal vision models when text extraction yields insufficient content.

PDF LLM Fallback: LLM processing is *not* the default. The routing hierarchy prioritizes deterministic parsers first (zero cost), then text-LLM for digital PDFs, and finally vision models?all are opt-in via separate install extras.

Current Version & Release Date: Version 0.0.8 was released on April 11, 2026 (per PyPI metadata).

24. assistant (2026-05-30T19:12:07.082Z)

I have enough verified detail now. Let me do one final fetch on the stmtforge repo directly (madhav921 is a downstream consumer; I need the actual source) to nail its license, and confirm pdfplumber's license/capabilities for the ranking baseline.

pdfplumber Overview

****License:**** MIT license

****Table Extraction Capabilities:**** Yes, pdfplumber supports borderless and semi-ruled tables through customizable strategies. The documentation notes: "vertical_strategy" and "horizontal_strategy" can be set to "lines," "lines_strict," "text," or "explicit" modes, allowing flexible detection of table boundaries using graphical lines or text alignment.

****Password-Protected PDFs:**** Supported. To load such files, "pass the `password` keyword argument, e.g., `pdfplumber.open("file.pdf", password = "test")`."

****Built On:**** pdfminer.six - the library is constructed on this foundation for parsing PDFs and analyzing object positioning.

****Active Maintenance:**** Yes. The latest release (v0.11.9) is dated January 5, 2026, confirming ongoing development.

****Pure Python with No Native Dependencies:**** The documentation indicates pdfplumber is pure Python. However, note that some alternative libraries like pymupdf require non-Python software installation, but pdfplumber itself avoids this requirement.

26. user (2026-05-30T19:12:17.764Z)

Web search results for query: "stmtforge PyPI package homepage source repository license MIT Apache"

Links: [{"title": "PyPI · The Python Package Index", "url": "https://pypi.org/"}], [{"title": "TestPyPI · The Python Package Index", "url": "https://test.pypi.org/"}], [{"title": "PyPI Package Registry | Forgejo ? Beyond coding. We forge.", "url": "https://forgejo.org/docs/latest/user/packages/pypi/"}], [{"title": "python-forge · PyPI", "url": "https://pypi.org/project/python-forge/"}], [{"title": "pyforge · PyPI", "url": "https://pypi.org/project/pyforge/"}], [{"title": "PyPI Package and Documentation Storage", "url": "https://files.pythonhosted.org/"}], [{"title": "PyPI packages in the package registry | GitLab Docs", "url": "https://docs.gitlab.com/user/packages/pypi_repository/"}], [{"title": "PyPI Download Stats", "url": "https://pypistats.org/"}], [{"title": "forge · PyPI", "url": "https://pypi.org/project/forge/"}], [{"title": "PyPI - The Python Package Index", "url": "https://github.com/pypi"}]

The initial search didn't return specific results for "stmtforge". Let me search more specifically for that package.

Links: [{"title": "How to Fork a Repository on GitHub | Learn Version Control with Git", "url": "https://www.git-tower.com/learn/git/faq/github-fork-repository"}], [{"title": "forge/docs/forge.org at main · magit/forge", "url": "https://github.com/magit/forge/blob/main/docs/forge.org"}], [{"title": "SourceForge Support / Documentation / Git", "url": "https://sourceforge.net/p/forge/documentation/Git/"}], [{"title": "Creating a repository from a template - GitHub Docs", "url": "https://docs.github.com/en/repositories/creating-and-managing-repositories/creating-a-repository-from-a-template"}], [{"title": "Fork a repository - GitHub Docs", "url": "https://docs.github.com/en/pull-requests/collaborating-with-pull-requests/working-with-forks/fork-a-repo"}], [{"title": "GitHub - PatNei/GITHUB2FORGEJO: Github 2 Forgejo: This script migrates your Github repositories to a Forgejo instance using the Forgejo API. It fetches your repositories from Github and automatically creates matching repositories on Forgejo. It's ideal for backup, migration, or maintaining a self-hosted mirror of your Github projects.", "url": "https://github.com/PatNei/GITHUB2FORGEJO"}], [{"title": "GitHub - Byte-Barn/gitcraft: Go to for Generic github templates · GitHub", "url": "https://github.com/RafaelJohn9/gitcraft"}], [{"title": "Cloning and forking repositories from GitHub Desktop - GitHub Docs", "url": "https://docs.github.com/en/desktop/adding-and-cloning-repositories/cloning-and-forking-repositories-from-github-desktop"}], [{"title": "GitHub · Change is constant. GitHub keeps you ahead. · GitHub", "url": "https://github.com/"}], [{"title": "GitHub - srstevenson/git-forge: Open

Git remotes in your web browser with a Git subcommand, supporting platforms like GitHub and GitLab. · GitHub", "url": "https://github.com/srstevenson/git-forge"]]

Let me search more specifically for "stmtforge" on PyPI.

Links: [{"title": "PyPI · The Python Package Index", "url": "https://pypi.org/"}, {"title": "Test PyPI", "url": "https://test.pypi.org/"}, {"title": "Client Challenge", "url": "https://pypi.org/project/tsforge/"}, {"title": "Help · PyPI", "url": "https://pypi.org/help/"}, {"title": "forge · PyPI", "url": "https://pypi.org/project/forge/"}, {"title": "Pypi", "url": "https://pypi.org/project/pipeforge/"}, {"title": "python-forge · PyPI", "url": "https://pypi.org/project/python-forge/"}, {"title": "pyforge · PyPI", "url": "https://pypi.org/project/pyforge/"}, {"title": "Using TestPyPI - Python Packaging User Guide", "url": "https://packaging.python.org/guides/using-testpypi/"}, {"title": "PyPI Package Registry | Forgejo ? Beyond coding. We forge.", "url": "https://forgejo.org/docs/latest/user/packages/pypi/"}]

The search didn't find a package specifically named "stmtforge". Let me try a direct search without quotes.

Links: [{"title": "GitHub - Rickaym/pyforge: The Python language provider for Minecraft Forge. · GitHub", "url": "https://github.com/Rickaym/pyforge"}, {"title": "7. Simple statements ? Python 3.14.5 documentation", "url": "https://docs.python.org/3/reference/simple_stmts.html"}, {"title": "python-forge · PyPI", "url": "https://pypi.org/project/python-forge/"}, {"title": "GitHub - materials-data-facility/forge: Forge is the Python package to access data in the Materials Data Facility", "url": "https://github.com/materials-data-facility/forge"}, {"title": "python:pyforge packages - Repology", "url": "https://repology.org/project/python:pyforge/packages"}, {"title": "pyforge · PyPI", "url": "https://pypi.org/project/pyforge/"}, {"title": "forge-python-sdk · PyPI", "url": "https://pypi.org/project/forge-python-sdk/"}, {"title": "Python Basics for Beginners | Print Statement, Variables & Comments | Learn Python from ScratchPART1 - YouTube", "url": "https://www.youtube.com/watch?v=0cV3RbRcTgY"}, {"title": "python-forge - conda-forge | Anaconda.org", "url": "https://anaconda.org/conda-forge/python-forge"}]

Based on my searches, I was unable to find a Python package specifically named "stmtforge" on PyPI or in other package repositories. The search results returned various "forge" packages (like `pyforge`, `python-forge`, `forge`, `forge-python-sdk`), but none specifically named "stmtforge".

It's possible that:

1. The package name might be spelled differently
2. It might be a very new or very niche package not yet widely indexed
3. It might be hosted in a private repository rather than on the public PyPI
4. The package name might be incomplete or misremembered

If you have more information about the package or where it's hosted, I'd be happy to search again. Alternatively, you could try searching directly on <https://pypi.org/> or checking if it exists in a GitHub repository with a different spelling or naming convention.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

27. assistant (2026-05-30T19:12:28.255Z)

I have comprehensive, verified findings. The stmtforge source repo couldn't be pinned to a canonical owner (it's referenced via downstream consumers like madhav921/Card-Statement-Analyser, and its PyPI page is behind a bot-challenge), so I'll flag that licensing as unverified. I have enough to deliver a ranked shortlist. Let me compile the structured output.

Key verified facts driving the ranking:

- **pdfplumber**: MIT, pure-Python (pdfminer.six), no native deps, password support via `password=` kwarg, semi-ruled table support via `text`/`lines` strategies, actively maintained (v0.11.9 Jan 2026). Best license fit.
- **Camelot (camelot-py)**: MIT, **v1.0.0+ switched default backend to pdfium (ships as wheel,

no system deps); Ghostscript now OPTIONAL**, OpenCV only for `engine="combined"`. Actively maintained (v1.0.9, Aug 2025). The old AGPL-Ghostscript concern is largely resolved if you stay on the pdfium backend. Note: the `pypdf\_table\_extraction` fork was archived/deprecated Apr 2025 ? do NOT use it; it points back to camelot-py.

- **Indian-specific prior art**: vishwaraja/hdfc-pdf-converter (HDFC savings, pdfplumber+camelot+PyPDF2, multi-line narration handling ? directly relevant but license unconfirmed); xaneem/hdfc-credit-card-statement-parser (MIT, CC only, has password support); MalayPalace/Bank-Statement-Utility (MIT, Python, config-driven HDFC/SBI/Kotak savings, uses pypdf, actively maintained to May 2026); stmtforge (Indian multi-bank CC, offline, license unverified).

28. user (2026-05-30T19:12:59.922Z)

Structured output provided successfully

29. assistant (2026-05-30T19:13:10.542Z)

I've completed the research and returned the ranked shortlist via StructuredOutput. Summary of the key findings for muneem's HDFC statement parsing:

License-critical correction: The most important finding is that **Camelot (camelot-py) v1.0.0+** switched its default backend to pdfium (a self-contained wheel), making Ghostscript/AGPL optional ? the old "avoid Camelot due to AGPL Ghostscript" guidance is now outdated if you stay on the pdfium backend. Separately, the **`pypdf\_table\_extraction`** fork was archived and deprecated in April 2025**, redirecting users back to camelot-py ? so do not adopt that fork.

Ranked shortlist (all permissive-licensed, local-capable):

1. **vishwaraja/hdfc-pdf-converter** (high) ? the only prior art specifically for HDFC *savings* statements; documents the multi-line-narration row-clustering heuristic. License unconfirmed in-page; verify before reuse.
2. **camelot-py** (high) ? MIT, pdfium default, no JVM, actively maintained (v1.0.9, Aug 2025).
3. **pdfplumber** (high) ? MIT, pure-Python, password support, semi-ruled tables via text/explicit strategies; the safest license fit and muneem's existing dependency.
4. **MalayPalace/Bank-Statement-Utility** (high) ? MIT, Python, config-driven per-bank parsers for HDFC/SBI/Kotak savings + CC; best architecture template.
5. **xaneem/hdfc-credit-card-statement-parser** (medium) ? MIT, HDFC CC only, has password + Forex handling.
6. **stmtforge** (medium) ? promising offline Indian multi-bank scope, but license/source unverified ? a lead to vet, not a vetted dependency.

Concrete recommendation for muneem: hand-roll coordinate/x-position clustering on pdfplumber (MIT, zero native deps, already in use), borrowing the multi-line-narration logic from vishwaraja/hdfc-pdf-converter, and keep camelot-py (pdfium backend) as a fallback. This avoids every AGPL/JVM trap while staying local-first.